Density

Exam code: 0625 & 0972

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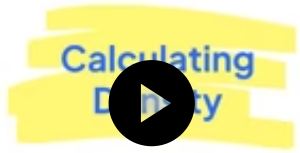

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Density

- Density is defined as:

The mass per unit volume of a material

- Density is related to mass and volume by the following equation:

$$\rho = \frac{m}{V}$$

- Where:
  - ρ = density, measured in kilograms per metre cubed (kg m^{-3})
  - m = mass, measured in kilograms (kg)
  - V = volume, measured in metres cubed (m^3)

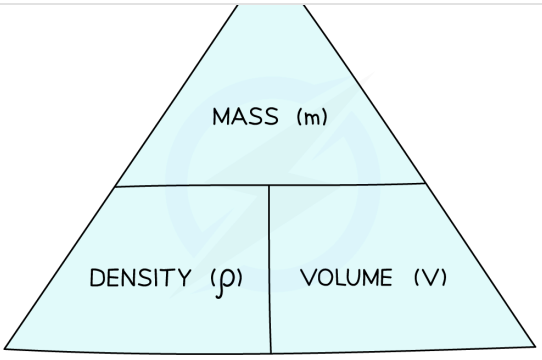
Formula triangle for density, mass and volume

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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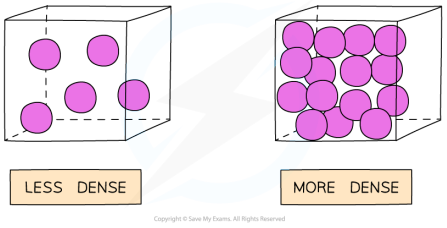
To use a formula triangle, simply cover up the quantity you wish calculate and the structure of the equation is revealed

- Objects made from **low density** materials typically have a **low mass**
- Similarly sized objects made from **high density** materials have a **high mass**
  - For example, a bag full of feathers is far lighter compared to the same bag full of metal
  - Or another example, a balloon is less dense than a small bar of lead despite occupying a larger volume
- Gases, for example, are generally less dense than solids because the particles in a gas are more spread out (same mass, over a larger volume)

Comparing the density of solids and gases



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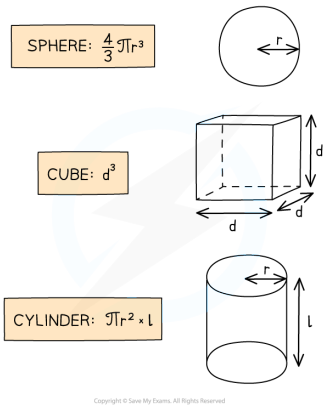
A gas is less dense than the same substance in liquid or solid form

- The units of density depend on what units are used for mass and volume:
  - If the mass is measured in g and volume in cm^3, then the density will be in g/cm^3
  - If the mass is measured in kg and volume in m^3, then the density will be in kg/m^3

Determining volume to calculate density

- The volume of an object may not always be given directly, but can be calculated with the appropriate equation depending on the object's shape



Volumes of common 3D shapes

Worked Example

A paving slab has a mass of 73 kg and dimensions 0.04 m × 0.5 m × 0.85 m.

Calculate the density, in kg/m^3, of the material from which the paving slab is made.

Answer:

Step 1: List the known quantities

- Mass of slab, m = 73 kg
- Volume of slab, V = 0.04 m × 0.5 m × 0.85 m = 0.017 m^3

Step 2: Write out the equation for density, mass and volume

$$\rho = \frac{m}{V}$$

Step 3: Substitute in values

$$\rho = \frac{73}{0.017}$$

$$\rho = 4294 \text{ kg/m}^3$$

Step 4: Round the answer to two significant figures

$$\rho = 4300 \text{ kg/m}^3$$

Examiner Tips and Tricks

Make sure you are comfortable converting between units such as metres (m) and centimetres (cm) or grams (g) and kilograms (kg).

- When you convert a **smaller** unit to a **larger** one, you **divide (÷)**
  - E.g. 5 g = 5 ÷ 1000 = 0.005 or 5×10^{-3} kg

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